# Session 8: Solutions and facilitation notes

Answer keys for `Course/Practicals/s8_exercises.md` (Session 8 exercise sheet), Parts A–C. Model answers for the appraisal (Part D) are in `Course/Resources/appraisal_instructor_guide.md`. Numbers in Part B are from `Course/Data/key_results.md` (SIMULATED data).

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## Part A: Writing exercises

### A1. Fix the spin (model rewrites)

| # | Model rewrite | Why |
|---|---|---|
| a | "The point estimate favoured the restrictive strategy, but the 95% CI was wide (−15.2 to 41.6 ml/kg) and included no difference; a larger trial would be needed to detect a smaller effect." | p = 0.42 is not a trend; report the estimate and its uncertainty (annotation #41). |
| b | "A positive fluid balance on the first ICU day was not associated with 28-day mortality (HR 1.17, 95% CI 0.85 to 1.60)." | "Impact" is causal; the cohort shows association only (#27). |
| c | "Positive cumulative fluid balance during the first three ICU days and 28-day mortality in sepsis: a propensity score-matched cohort study" | Title must match the observational design (#1). |
| d | "To our knowledge, this is among the lowest day-5 fluid balances reported in septic shock trials [refs]", or delete the sentence. | Superlatives need protection and a citation (#40). |

### A2. Verb ladder

a: **was observed in** (case series) · b: **is associated with** (cohort) · c: **did not reduce** (one RCT; add "in this pilot" because it is small) · d: **reduces / causes** (consistent large RCTs and meta-analysis) · e: **was observed / was … %** (descriptive cross-sectional).

### A3. Results sentences

1. "In the Day-3 matched cohort, a positive cumulative fluid balance was associated with higher 28-day mortality (HR 1.56, 95% CI 1.10 to 2.22; P = 0.012)."
2. "By day 28, 6 of 24 patients had died in each group (relative risk 1.0, 95% CI 0.38 to 2.66)." In the Discussion: "the trial was too small to assess mortality".
3. "In the Day-1 cohort, a positive fluid balance was not associated with 28-day mortality (HR 1.17, 95% CI 0.85 to 1.60; P = 0.354)." Do not write "had no effect".

### A4. Title from PICO (running example)

"Restrictive vs usual fluids and 28-day mortality in septic shock: a randomised controlled trial" (14 words). Or, for an observational proposal: "Cumulative fluid balance and 28-day mortality in adults

with septic shock: a retrospective cohort study". Check: 10–16 words, design named, no "impact", observational = "and" / "association between".

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# Part B: Manuscript workshop 2: improving the Results, and models for the other sections

## Improving the class Q1 Results (part 1 of the workshop)

| Check | Typical fix in the class draft |
|---|---|
| Every number matches results pack Q1 | Correct it from the pack, e.g. 134/373 (35.9%) |
| Percentages with denominators | "35.9%" becomes "134 of 373 (35.9%)" |
| Crude and adjusted both given | Crude OR 2.49 (1.70 to 3.70), adjusted 1.62 (1.02 to 2.59), n = 527 |
| Tables stand alone | Title with who and when, n per column, units, footnote for SOFA and IQR |
| No interpretation | Delete "because sicker patients received more fluid" |
| Working title | "Positive fluid balance by day 3 and 28-day mortality in sepsis: a multicentre cohort study" |

The models below are for the Introduction, Discussion and Abstract taught in 8.1. They are optional to submit. Use them to steer, do not paste them. Participants who chose Q2–Q5 follow the same structure with their own pack.

All numbers from `Course/Data/key_results.md` and results pack Q1 (**SIMULATED DATA**). References are placeholders [n]; participants use their Session 1–2 references and Papers 1–2.

## Model title and keywords

**Positive fluid balance by day 3 and 28-day mortality in sepsis: a multicentre cohort study** (15 words; design named; calibrated).

Keywords: Fluid Therapy; Water-Electrolyte Balance; Critical Care; Mortality; Cohort Studies.

## Model Introduction (~300 words, three paragraphs)

1. *Known.* Intravenous fluids are a cornerstone of early sepsis treatment, but fluid given beyond the resuscitation phase accumulates. A positive cumulative fluid balance has been associated with higher mortality in several observational studies, including a large propensity-matched cohort in which a positive balance on ICU day 2 was associated with higher 28-day mortality (HR 2.13) [Paper 2].
2. *Not known.* Because sicker patients receive more fluid, it remains uncertain how much of this association reflects illness severity (confounding by indication) rather than fluid itself [refs]. Randomised trials of restrictive strategies, such as a recent pilot targeting non-resuscitative fluids [Paper 1], have been small and inconclusive.
3. *Aim.* We therefore examined whether a positive cumulative fluid balance by ICU day 3 was associated with 28-day mortality in adults with sepsis, before and after adjustment for severity.

## Model Discussion (~500 words, six moves)

1. **Answer.** In this multicentre cohort, a positive fluid balance by ICU day 3 was associated with higher 28-day mortality after adjustment for severity (adjusted OR 1.62, 95% CI 1.02 to 2.59).
2. **Compare.** The direction agrees with Paper 2 (day-2 HR 2.13, 95% CI 1.48 to 3.06), although our adjusted estimate was smaller; differences in case mix, timing of the balance and adjustment sets may explain this.
3. **Explain.** Most of the crude association (OR 2.49) disappeared after adjustment, consistent with confounding by indication: patients with a positive balance were sicker (median SOFA 8 vs 6; vasopressors 55.0% vs 24.9%). Fluid accumulation may contribute to organ oedema, but our data cannot show this.
4. **Limitations.** Residual confounding by unmeasured severity is likely and would exaggerate the association; reverse causation (patients who deteriorate accumulate fluid) acts in the same direction. Patients who died before day 3 had less time to accumulate fluid. The adjusted Cox HR (1.46, 95% CI 1.00 to 2.13) had a lower limit at 1. Lactate was missing in 15%, but multiple imputation gave a similar estimate (OR 1.70). The association looked stronger in heart failure (adjusted OR 4.39 vs 1.29; p for interaction 0.031) but was based on 90 patients and should be treated as a hypothesis. The data are simulated.
5. **Implications.** These findings do not justify restricting fluid in clinical practice; a randomised trial of fluid strategies after the resuscitation phase is needed.
6. **Conclusion.** A positive cumulative fluid balance by ICU day 3 was modestly associated with higher 28-day mortality; most of the crude association reflected severity of illness.

## Model abstract (≈190 words)

**Background:** A positive fluid balance is associated with death in sepsis, but sicker patients receive more fluid. We examined the association after adjustment for severity.

**Methods:** Multicentre prospective cohort (simulated teaching data) of 620 adults with sepsis in four ICUs, 2023–2024. Exposure: positive cumulative fluid balance by ICU day 3. Outcome: 28-day mortality. Logistic and Cox regression adjusted for age, SOFA, lactate, vasopressors, mechanical ventilation and chronic kidney disease; multiple imputation for missing lactate.

**Results:** Of 618 patients analysed, 373 (60.4%) had a positive balance; 134/373 (35.9%) vs 45/245 (18.4%) died. The crude OR was 2.49 (95% CI 1.70 to 3.70) and the adjusted OR 1.62 (95% CI 1.02 to 2.59); with imputation 1.70 (1.09 to 2.64). The adjusted HR was 1.46 (95% CI 1.00 to 2.13).

**Conclusions:** A positive fluid balance by day 3 was modestly associated with 28-day mortality; most of the crude association reflected illness severity. A randomised trial is needed before fluid strategies are changed.

## Question-specific Discussion points (from the packs)

| Question | Answer sentence (numbers from key_results.md) | Key limitation |
|---|---|---|
| Q2 AKI | Adjusted OR 1.27 per 10 ml/kg (95% CI 1.16 to 1.40) | Reverse causation: oliguric AKI causes fluid accumulation |
| Q3 VFD | Adjusted −0.61 days per 10 ml/kg (95% CI −1.00 to −0.21) | VFD combines death and ventilation; skewed outcome |
| Q4 Lactate | Adjusted OR per mmol/L 1.39 (CC, n = 527) vs 1.45 (MI, n = 620) | Missing not at random cannot be excluded; MAR plausible (missing in less sick patients) |

| Question | Answer sentence (numbers from key_results.md) | Key limitation |
|---|---|---|
| Q5 ICU stay | Adjusted ratio of geometric means 1.24 (95% CI 1.10 to 1.39), survivors (n = 366 adjusted) | Survivors only (conditioning on survival); skewed outcome |

# Part C: Getting published

## C1. Spot the predatory journal

**1. Red flags in the invitation (at least 8):**

1. Generic flattering greeting ("Respected Professor/Doctor"), unsolicited mass e-mail.
2. Free e-mail address, not a publisher domain.
3. Vague claim to have read "your excellent article" (no title).
4. Absurdly broad scope (medicine, nursing, engineering, management; "all areas of science").
5. A fake metric ("International Universal Index" impact factor). Only Clarivate issues the Journal Impact Factor.
6. Guaranteed review in 48 h and publication in 5 days.
7. Fees hidden until after acceptance.
8. Pressure tactics ("this week only"), multiple exclamation marks, unusual grammar.
9. Submission by replying to an e-mail instead of an editorial system.

**2. Journal profiles:**

| Journal | Verdict | Reason |
|---|---|---|
| J1 | **Trust** | Indexing verified at source, transparent fees and timelines, verifiable editors, uses reporting checklists. |
| J2 | **Avoid** | Google Scholar / Index Copernicus are not quality indexes; 72-h acceptance; hidden fees; unverifiable board. |
| J3 | **Trust / check further** | DOAJ listing and clear policies are good signs; no impact factor is normal for a new or small journal. Check it publishes work like yours. |
| J4 | **Avoid** | A "hijacked" or look-alike journal. Always reach the real journal through its publisher's website or the NLM Catalog, not through an e-mail link. |

## C2. Responding to reviewers (model rows)

| # | Model response | Change |
|---|---|---|
| R1.1 | "Thank you. We agree that our design supports association, not causation. We have changed the title to '… and …: a cohort study' and replaced 'improves' with 'was associated with' throughout." | Title p.1; abstract; Discussion p.x |
| R1.2 | "Thank you. The main result is now reported as RR 0.78 (95% CI 0.61 to 0.99) in the abstract, Results and Table 2." | Abstract; Results p.x |
| R1.3 | "Thank you for spotting this. 212 patients were included and 4 were excluded from the adjusted model because of missing lactate; the flow diagram and text now show 212 included and 208 analysed, with reasons." | Fig. 1; Results p.x |
| R2.1 | (polite rebuttal) "We agree 1-year mortality is important. Our protocol, approved by the ethics committee, followed patients to day 28 only, so these data were not collected. We now mention this as a limitation and suggest it for future studies." | Limitations p.x |

Marking points: every comment answered; thanks without grovelling; concrete location of every change; no new data invented; disagreement supported by a reason (protocol, ethics approval).